

Instacart Market Basket Analysis

High-Level Design (HLD)

Overview

This project takes raw grocery order data through cleaning, exploratory analysis, and into a business-facing report. Two data access strategies were built and compared early on: reading through a MySQL database via SQLAlchemy, and reading the source CSV files directly with pandas. The direct-read path was measured to be substantially faster and is the active path used for all analysis; the database ingestion path was kept in the codebase as a reusable capability.

System Flow

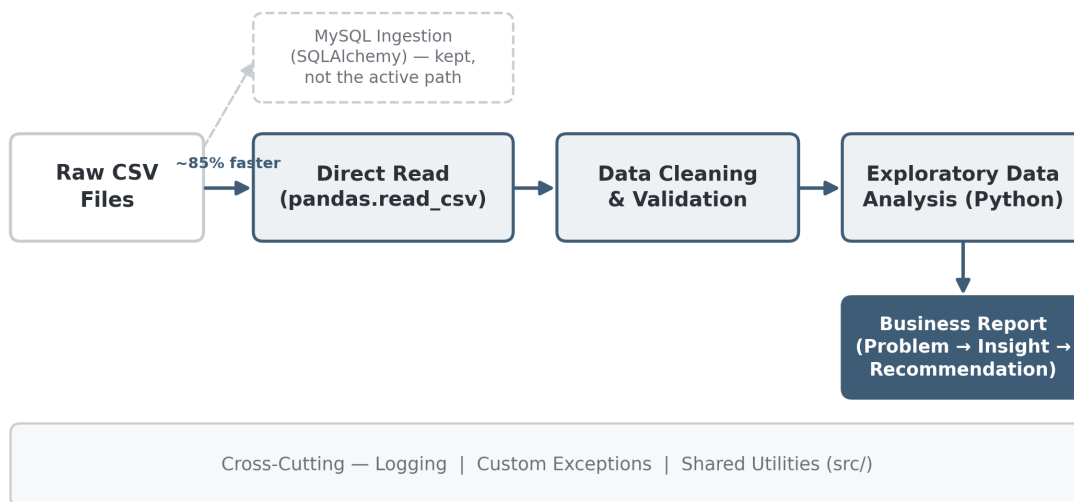


Figure 1. High-level data flow from raw files to the business report.

Components

Component	Responsibility
Ingestion (kept, inactive)	Loads raw CSVs into MySQL via a SQLAlchemy-managed engine.
Data Access (active)	Reads source CSVs directly from local storage for all analysis.
Cleaning & Validation	Null checks, data type validation, duplicate checks, referential integrity.
Exploratory Analysis	Python-based EDA across five business-question sections (pandas, seaborn).
Reporting	Consolidates findings into a business-facing report: problem, insight, recommendation.
Cross-Cutting	Logging and custom exception handling used throughout ingestion and processing.

Technology Stack

- Python (pandas, seaborn, matplotlib) for data processing and analysis.
- MySQL and SQLAlchemy for the database ingestion path.
- Jupyter Notebook as the primary analysis environment.